BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI, PILANI

FIRST SEMESTER 2023– 2024

CE F231 (FLUID MECHANICS) EVALUATIVE-5 (OPEN BOOK)

Duration: 45 mins **Max Marks**: 20

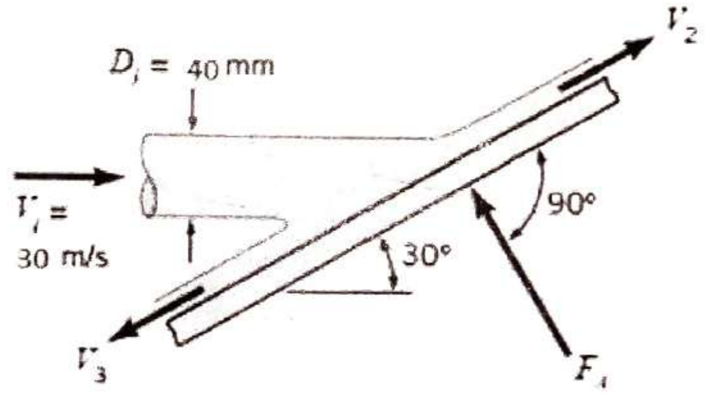
Note: Please write the answers in SI units and up to three decimal points with units. Use g=9.81 m/s^2.

Q1. A closed tank partially filled with water upto a height of 3 m, having an orifice of diameter 15 mm at the bottom of the tank. Determine the pressure required for a discharge of 5 litres/s through the orifice. Take coefficient of discharge is 0.65. Express the pressure in Pascal. **[5]**

Q2. A cylindrical tank is having a hemispherical base. The height of cylindrical portion is 5 m and diameter is 4 m. At the bottom of this tank is fitted with an orifice of diameter 200 mm is fitted. Find the time required to completely emptying the tank. Take coefficient of discharge 0.62. **[5]**

Q3. A horizontal jet of air strikes a stationary flat plate as indicated in figure below. The jet velocity is 30 m/s and jet diameter is 40 mm. If the air velocity magnitude remains constant as the air flows over the plate surface in the directions shown. Determine:

a) the magnitude of F_A the anchoring force required to hold the plate stationary
b) the fraction of mass flow along the plate surface in each of the two directions shown
c) the magnitude of F_A the anchoring force required to allow the plate to move to the right at a constant speed of 5 m/s



[10]